State variables of a real gas

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Properties of real gases are studied by measuring different isotherms of SF_6 between the temperatures 31.0 °C and 49.8 °C. The critical coefficient of SF_6 was calculated to be $K_\mathrm{k} = (2.79 \pm 0.65)$, the vaporization energy to $L = (142 \pm 19)\,\mathrm{kJ/kg}$, the critical temperature to $T_\mathrm{K} = (47.2 \pm 2.0)\,°\mathrm{C}$ and the coefficients a and b of the van der Waals-equation to $a = (0.820 \pm 0.046)\,\mathrm{Pa\,m^6/mol^2}$ and $b = (91.2 \pm 5.0) \times 10^{-6}\,\mathrm{m^3/mol}$. The van der Waals-equation with these parameters was compared to the experimental data for the isotherm at 31.0 °C. The critical volume was $V_\mathrm{K} = (0.55 \pm 0.05)\,\mathrm{ml}$, the critical pressure $p_\mathrm{K} = (36.5 \pm 2.0)\,\mathrm{bar}$.

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1 Introduction

An ideal gas is modeled with point molecules which do not interact, except in collision. Therefore, the ideal gas equation that describes its behavior has a limited range of validity. The denser the gas is, the higher its pressure and the lower its temperature, the more deviation from this simplified model is observed. Due to the fact, that real gases don't just consist of point molecules and thus do interact in more different ways than collision, a more accurate theory is needed. This led to theoretical models that include more than eight constants inside the equation of state. Of course, it comes with a high level of complexity.
In this experiment, we will examine how close a first approximation can get by studying the van der Waals-model, which consists of two parameters a and b that take into account the attractive and the repulsive forces of the molecules. We will use a SF_6-gas and compare the results with the theoretical model.

2 Theoretical background

If a liquid is introduced into a vessel which has previously been completely evacuated and which only partially fills the vessel, part of the liquid evaporates and a pressure characteristic of the liquid is established above it, which is referred to as its saturation vapor pressure. By lowering the volume of the vessel at constant temperature, the pressure does not change. A part of the vapor changes to the liquid state. If the volume is increased at constant temperature, parts of the liquid state changes into vapor. Only when all the liquid has evaporated with an increase in volume, the pressure decreases and the properties of the vapor approach more and more those of ideal gases. The liquid state and the vaporous state are called "phases" of the same substance. At the saturation vapor pressure, the liquid phase and vaporous phase are in equilibrium. For a molecule to leave the liquid phase, it has to overcome the attractive forces inside the liquid. When a particle in the vapor phase hits the surface, it enters the liquid phase. Therefore equilibrium state can be understood in terms of equal rates of transition between particles of both phases [2].

The probability of a particle having the energy that is needed to leave the liquid phase is described by the Boltzmann-distribution. Only at the (T, p)- values on the vapor pressure curve, the two phases can coexist. This curve thus divides the (T, p) plane into two areas, as seen in fig. 1: above, the substance can only exist in the liquid, below, it can only exist in the gaseous state. It is visible that the saturation vapor increases with increasing temperature T. In [2], the vapor pressure curve is described with:

$$p_\mathrm{v} = Ae^{-\frac{E}{k_\mathrm{B}T}} \propto e^{-\frac{\Lambda}{RT}} \tag{1}$$

where E is the evaporation energy per molecule, Λ the molar evaporation energy and N_A is Avogadro's constant $N_\mathrm{A} = 6.02214076 \times 10^{23}\,1/\mathrm{mol}$. $R = N_\mathrm{A} k_\mathrm{B}$ is the ideal gas constant ($R = 8.314462618\,\mathrm{J/Kmol}$) whereas k_B the Boltzmann constant ($k_\mathrm{B} = 1.380649 \times 10^{-23}\,\mathrm{J/K}$) [3].

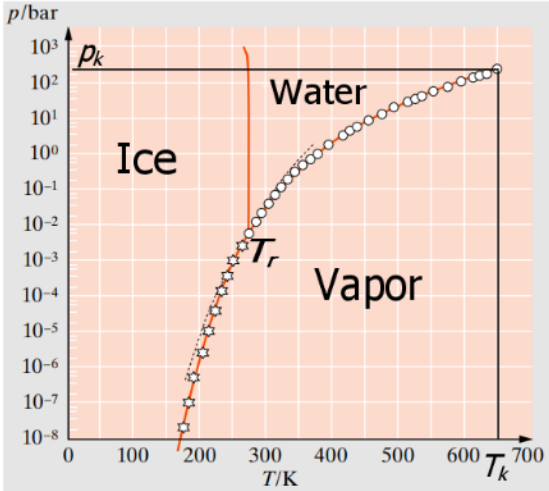

Fig. 1: The diagram depicts the vapor pressure curve (round dots) for water with semi-logarithmic axes. Above the triple point T_r, it divides the (T, p)-plane into two parts and ends with the critical temperature T_k and the critical pressure p_k. Only for points on this curve, vapor and liquid can coexist. For lower temperatures beneath T_r, it is shown the sublimation curve (star shaped points). The vertical line starting from T_r is the melting curve. Figure Adapted from [2].

The factor A can also depend marginally on the temperature. Therefore, as visible in fig. 1 the vapor pressure curve is not perfectly linear in a semi-logarithmic $p(T)$-diagram.

In our experiment we measured ten isotherms for the gas sulfur-hexafluoride (SF_6), by measuring the pressure of the gas while varying the volume successively. Theoretically, the curve should take the path as predicted by the van der Waals-equation [1]:

$$(p + \frac{\nu^2 a}{V^2})(V - \nu b) = \nu RT \tag{2}$$

Where p is the pressure, V the volume, ν the quantity of material of the considered gas. a and b are positive constants. In general, for different temperatures we expect a band of curves in fig. 2, where it was calculated few isotherms for CO_2.
It is visible, that the theoretical curve $(ABCDE)$ does not exactly match the shape it would have in a real measurement, indicated with the line (ACE), because for temperatures lower than the critical temperature T_k, liquid and vapor are distinguishable and thus an equilibrium between the two phases is formed. As a result, the pressure stays constant in a certain interval of volume. The real isotherms are matched with the Maxwell construction, which connects the points A and E with a constant level, so that the areas above and below this horizontal line are equal. The pressure at which one isotherm reaches its plateau depends on its temperature and thus follows the vapor pressure curve as shown in fig. 1.

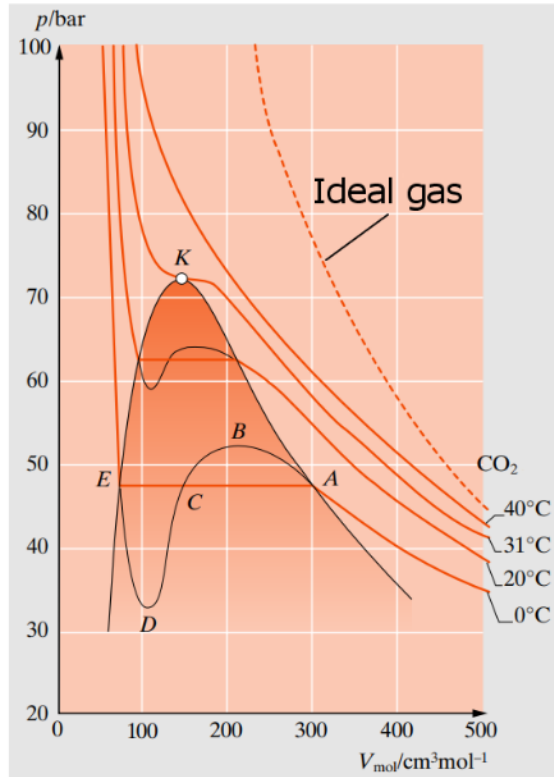

Fig. 2: The theoretical isotherms for 1 mol CO_2-gas and an ideal gas are shown. Depending on the temperature of a real gas, a plateau (Point A to E) is formed where the pressure stays constant within an interval of volume. Above the critical temperature T_k, the isotherms exhibit no plateau anymore. Adapted from [2].

At higher temperatures above T_k there is no defined transition between gas and liquid, and thus the isotherms gradually approach the hyperbolic shape of ideal gases. In fig. 2 this critical point is marked with $K = (V_\mathrm{k}, p_\mathrm{k})$, where V_k is the critical volume and p_k the critical pressure. By placing the conditions $\frac{dp}{dV} = 0$ and $\frac{d^2p}{dV^2} = 0$ on the van der Waals-equation, the critical values follow with:

$$p_\mathrm{k} = \frac{a}{27b^2} \tag{3}$$

$$V_\mathrm{k} = 3b \tag{4}$$

$$T_\mathrm{k} = \frac{8a}{27Rb} \tag{5}$$

The exact derivation is added in the appendix. With the following experiment we measured the isotherms of a real SF_6-gas for various temperatures to determine these critical values.

3 Measuring isotherms of SF_6

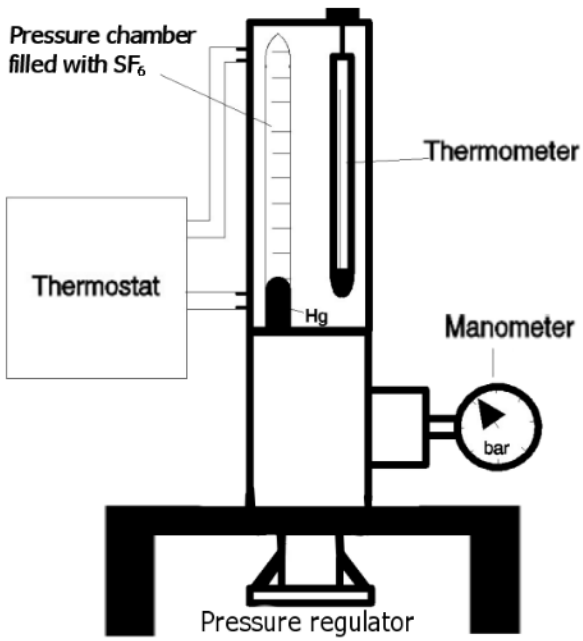

Fig. 3: A schematic picture of the measuring apparatus. Mercury is filled in the glass column and above it the SF_6-gas which we analyzed. The pressure can be regulated with the wheel. The temperature is measured with the Voltcraft DT 300 and as thermostat we used the LAUDA 104. Adapted from [1].

to measure the isotherms for different temperatures, we measured the pressure depending on the volume, starting with high volume. At the right edge of the plateau we then measured more values to observe the exact point, at which liquid and vapor phases first reach an equilibrium state. For lower temperatures, we took more data points on the left edge as well to see the sudden change of pressure that indicates the end of equilibrium between the liquid and the vaporous phase. For the error of reading it was chosen the half of a scale division. That is $\sigma_\mathrm{V} = 0.025\,\mathrm{ml}$ and $\sigma_\mathrm{p} = 0.25\,\mathrm{bar}$. The temperature error of $\sigma_\mathrm{T} = 2\,°\mathrm{C}$ is given in the device description of the used Voltcraft DT 300. Fig. 4 shows the measured isotherms for 10 different temperatures, starting with 31 °C and increasing up to 49.8 °C. There the temperature errors originate from the range of variation of the digital read-off during the measurement. While measuring each isotherm, the temperature almost stayed constant. The global temperature error of σ_T which is larger, just gives an equally unknown uncertainty for every data point in the diagram and thus is not considered at first.

We then chose the first isotherm that did not exhibit a discrete plateau anymore. In our case, it was at the isotherm for $T = (47.0 \pm 0.1)\,°\mathrm{C}$, since

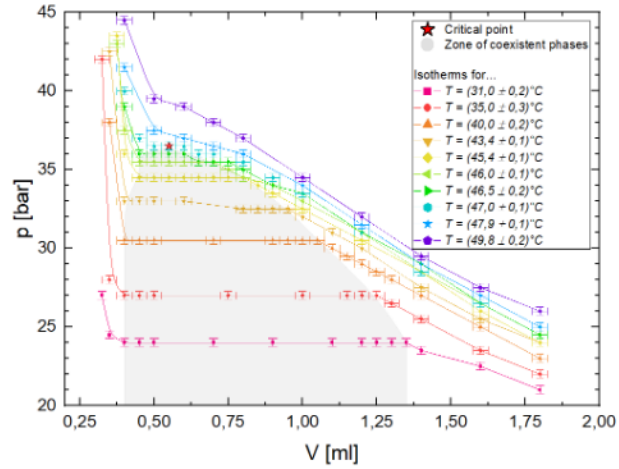

Fig. 4: Graphic of the ten measured isotherms for different temperatures. The grey zone marks the area of coexistent phases of liquid and vapor. The critical point K lies on the $T = (47.0 \pm 0.1)\,°\mathrm{C}$-isotherm. The critical point was extrapolated and is marked with a red star at $V_\mathrm{k} = (0.55 \pm 0.05)\,\mathrm{ml}$ and $p_\mathrm{k} = (36.5 \pm 0.2)\,\mathrm{bar}$.

the pressure was constant for a volume interval of $\Delta V = 0.1\,\mathrm{ml}$. Above that temperature, no visible plateau occurred anymore. Therefore we extrapolated the critical volume and the critical pressure to be at the middle of this last visible plateau:

$$\begin{aligned} p_\mathrm{k} &= (36.5 \pm 2.0)\,\mathrm{bar} \\ V_\mathrm{k} &= (0.55 \pm 0.05)\,\mathrm{ml} \end{aligned} \tag{6}$$

Technically it is possible to extrapolate the critical pressure with higher precision. The temperature error of 2 °C caused by the thermometer Voltcraft DT 300 (which consequently is inherent to every isotherm) outweighs the error of reading, though. The temperature error then propagated into the error of pressure, that is chosen to be 2 bar. The chosen error for the critical volume is determined by the distance to the next data points left and right, which is $\sigma_{\mathrm{V,\,k}} = 0.05\,\mathrm{ml}$. The extrapolated critical pressure agrees with the literature value of $p_\mathrm{k} = 37.49\,\mathrm{bar}$ [4].

4 Evaluating the critical temperature and the latent heat of evaporation

The pressure level, at which each isotherm reaches its plateau depends on the temperature. As discussed in section 2 this level follows the vapor pressure curve p_v and thus is described with an exponential law. By plotting the pressure p_v of each isotherms plateau logarithmic against the inverse of its temperature T, a

linear correlation should be observable, according to eq. (1), which has a slope that is:

$$\eta = \frac{\Lambda}{R} \tag{7}$$

This equation can be used to evaluate the molar evaporation energy Λ, and, by knowing the molar mass M_{mol} of the considered gas, the heat of evaporation (or latent heat of evaporation) L_{evap},

$$L_{\text{evap}} = \frac{\Lambda}{M_{\text{mol}}}. \tag{8}$$

The vapor pressure diagram for the measured isotherms can be seen in fig. 5. The isotherms measured at $T > 47,0\,°\text{C}$ were excluded since they showed no clear pressure plateau.

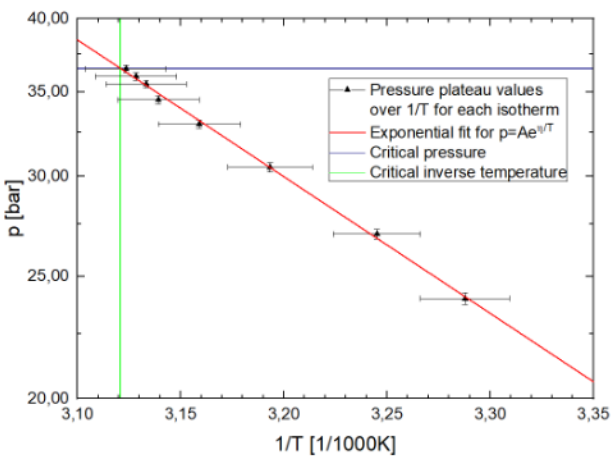

Fig. 5: This semilogarithmic diagram shows the pressure plateau values plotted over $1/T$ for the isotherms with $T \leq 47.0\,°\text{C}$. An exponential fit for $Ae^{-\frac{\eta}{T}}$ was applied so that eq. (7) can be used to determine the molar evaporation energy Λ. The blue line marks the critical pressure $p_{\text{k}} = (36.5 \pm 2.0)$ bar. The green line highlights the critical inverse temperature.

On our data was applied an exponential fit for $Ae^{-\frac{\eta}{T}}$, where A is a positive constant. The slope of this linear correlation that is visible with a logarithmic pressure scale was determined to be $\eta = (2.50 \pm 0.33) \times 10^3$ K. To extrapolate the critical temperature, it was drawn a horizontal line (blue) at the critical pressure level. By evaluating the crossing point with the exponential vapor pressure curve-fit this led to an inverse temperature of $1/T = (3.121 \pm 0.020)\,1/1000\text{K}$. for the crossing point error was chosen the same error as of the lowest inverse temperature value. With that, the critical Temperature follows as:

$$T_{\text{k}} = (47.2 \pm 2.0)\,°\text{C}$$

Within the range of uncertainty, this value is in agreement with the literature value for the critical temperature that is $T_{\text{k}} = (45.6)\,°\text{C}$ [5]. Considering the slope η, the molar evaporation energy follows with eq. (7):

$$\Lambda = (20,7 \pm 2,7)\,\frac{\text{kJ}}{\text{mol}}$$

SF_6 has a molar mass of $M_{\text{mol}} = 146,05\,\text{g/mol}$ [5]. With eq. (8) this results in a latent heat of evaporation:

$$L_{\text{evap}} = (142 \pm 19)\,\frac{\text{kJ}}{\text{kg}}$$

The evaluated heat of evaporation is in agreement with the literature value of $L_{\text{evap, ref.}} = 153,2\,\text{kJ/kg}$ [6].

5 Chemical amount of SF_6

The chemical amount of a substance is the number of contained particles divided by Avogadro's constant. With this quantity, the ideal gas equation (cf. eq. (17)) is expressed. In this model, the particles are assumed to be monoatomic and non-interacting with each other. Thus, the ideal gas approximation is better for smaller particle numbers per volume N/V, because in this case less interactions such as scattering actually occur. This trend can also be seen in fig. 8. To determine the chemical amount of SF_6 that was being used for this experiment, we exemplary picked the isotherm for $T = 40\,°\text{C}$ and computed analytically the intersection of a linear fit through the data with the y-axis in a $pV(1/V)$-diagram, seen in fig. 6. For this, only the last five points were taken into account, because this way a good compromise between enough data for a fit and high enough volume was reached.

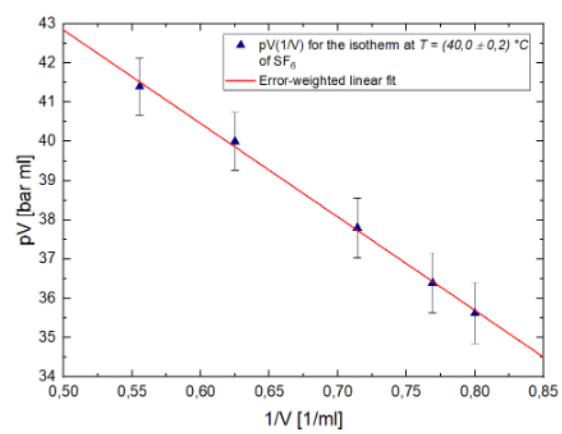

Fig. 6: $pV(1/V)$ curve of SF_6 at $40\,°\text{C}$. Only the data with high inverse volume $1/V_{\text{max}} > 0.50$ 1/ml was taken into account. The chemical amount of SF_6 is proportional to the fit at $1/V = 0$, where the van der Waals-equation becomes equal to the ideal gas equation.

For $V \to \infty$, it can be shown that the van der Waals-equation eq. (2) equates the ideal gas equation eq. (17). Hence we interpolated the linear fit to find its intersection ξ with the pV-axis at

$$\xi = (5.47 \pm 0.27)\,\mathrm{Pa} \cdot \mathrm{m}^3.$$

The ideal gas equation then becomes $\nu = \xi/RT$, thus for $T = (313.2 \pm 0.2)\,\mathrm{K}$ the chemical amount is equal to

$$\nu = (2.10 \pm 0.43) \times 10^{-3}\,\mathrm{mol}$$

It has to be added that this value is only an approximation because the range of volume used to determined it was obviously finite, violating the made assumption that the van der Waals equation is the same as the ideal gas equation.

6 Determination of gas-specific parameters

The van der Waals equation eq. (2) contains two parameters, the internal pressure a/V^2 and the covolume b. Both are specific for every gas and can only be found experimentally. With knowledge of the critical volume V_k, pressure p_k and temperature T_k, the eqs. (14), (16) and (15) in the appendix can be solved analytically for a and b:

$$a = \frac{27R^2\,T_\mathrm{k}^2}{64p_\mathrm{k}},$$
$$b = \frac{RT_\mathrm{k}}{8p_\mathrm{k}}$$

Also, it makes sense to define a critical coefficient K_k to measure the quality of the approximation of the van der Waals-model:

$$K_\mathrm{k} = R\,\frac{T_\mathrm{k}}{p_\mathrm{k}V_\mathrm{k}} \tag{9}$$

For the ideal gas, $K_\mathrm{k} = 1$. Since eq. (9) is independent of a and b, the model predicts for all real gases $K_\mathrm{k} > 8/3 \approx 2.67$. The further away from 8/3 the critical coefficient is for real experimental data, the worse the van der Waals-model can describe it.
The critical volume and critical pressure were already determined in chapter 3 and the critical temperature was calculated in chapter 4. Plugging these quantities in yields

$$a = (0.820 \pm 0.046)\,\mathrm{Pa\,m^6/mol^2},$$
$$b = (91.2 \pm 5.0) \times 10^{-6}\,\mathrm{m^3/mol} \quad \text{and}$$
$$K_\mathrm{k} = (2.79 \pm 0.65)$$

Reference values are provided by [4]: $a_\mathrm{lit.} = 0.786\,\mathrm{Pa\,m^6/mol^2}$, $b_\mathrm{lit.} = 88.0 \times 10^{-6}\,\mathrm{m^3/mol}$. The obtained parameters a and b can both be aligned with literature within the margin of error.

The critical coefficient is a function of the chemical amount of substance because the critical volume too depends on it. The greater the chemical amount, the greater is the critical volume. To acquire an approximate value of reference for the critical coefficient, the literature values for the critical temperature and critical pressure mentioned in chapter 3 together with our result for the critical volume and chemical amount (cf. chapter 5) were plugged into eq. (9). There, V_k was replaced by the inverse chemical particle density χ_c ($[\chi_\mathrm{c}] = \mathrm{m^3/mol}$) due to units, resulting in $K_{\mathrm{k,\,ref.}} = 2.69$. Our result agrees with this semi-literature value; this is partly because for the calculation of them some of the used quantities were the same, though. Our result also agrees with 8/3, the value the van der Waals equation predicts for all real gases. We thus conclude it is a solid approximation for the gas SF_6.

7 Comparison of experiment and theory

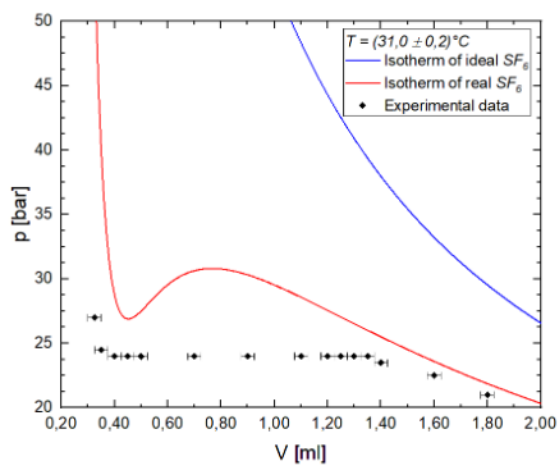

Fig. 7: The isotherms of SF_6 in the ideal gas model (blue), the van der Waals-model (red) and experimental data for $\nu = (2,10 \pm 0,43) \times 10^{-3}\,\mathrm{mol}$. The experimental data lies permanently below the van der Waals prediction and its curve deviates especially between $V \approx 0,40\,\mathrm{ml}$ and $V \approx 1,40\,\mathrm{ml}$.

With the parameters a and b that were experimentally determined in chapter 6, we are able to draw a theoretically expected isotherm of SF_6 in the frame of the van der Waals-model. Analogous to the procedure described in the appendix chapter 9, it is also possible to draw an isotherm for 'ideal' sulfur hexafluoride. Both isotherms at $T = 31\,°\mathrm{C}$ are drawn

in fig. 7 together with experimental data collected in chapter 3.

It is visible that the real gas curve, calculated with the van der Waals equation (2), resembles the theoretically expected shape, as exemplary seen in fig. 2 (whereas this figure shows isotherms for 1 mol of CO_2). Unexpected is the permanent overestimation of the experimental data by the van der Waals-model that is clearly visible in fig. 7. The measured critical pressure-plateau does not correspond to the theoretical Maxwell construction for the real gas, because the 'platform' for the experimental data lies below the construction line. Yet, the measured interval of volume, in which the pressure stays constant, is the same as theoretically expected.

We conclude that for sulfur hexafluoride the van der Waals-model predicts the right interval of volume for the Maxwell construction but a too high value for the pressure to be aligned with our measurement. Outside of the zone of coexistent phases the model is able to describe the curve of the experimental data well, except for a positive offset. This gives rise to the necessity of better, more accurate models that are able to describe the behavior of real gases with more precision. This of course comes with a loss of simplicity. For most applications, the van der Waals model provides a good compromise between accuracy and ease of use.

8 Conclusion

In this experiment it was studied the properties of a real gas, exemplified with SF_6. The isotherms for ten temperatures within the interval from $T = 30\,°\mathrm{C}$ to $T = 50\,°\mathrm{C}$ were measured and the critical pressure was determined to be $p_\mathrm{k} = (36,5 \pm 2,0)$ bar. The critical Volume was evaluated to be $V_\mathrm{k} = (0,55 \pm 0,05)$ ml. By plotting the pressure plateau of each isotherm over the inverse of the related isotherms temperature, the critical temperature was determined to be $T_\mathrm{k} = (47,2 \pm 2,0)\,°\mathrm{C}$. The critical pressure and the critical temperature are in agreement with the literature values [5]. With the slope η of an exponential fit on the vapor pressure curve the latent heat of evaporation was determined to $L_\mathrm{evap} = (142 \pm 19)\,\mathrm{kJ/kg}$ and agrees with the literature value in [6]. The molar evaporation energy was also computed to be $\Lambda = (20,7 \pm 2,7)\,\mathrm{kJ/mol}$. The quantity of material was extrapolated by plotting the product pV over $1/V$ for large volumes of the isotherm at $T = 40\,°\mathrm{C}$. The intersection of the best fit with the pV-axis was used to interpolate the volume into infinity, because for $1/V \to 0$, the van der Waals gas becomes and ideal gas . Then the ideal gas equation was used to determine the quantity of material to be $\nu = (2,10 \pm 0,43) \times 10^{-3}\,\mathrm{mol}$. Finally the parameters within the van der Waals equations were evaluated to be $a = (0,820 \pm 0,046)\,\mathrm{Pa\ m^3/mol^2}$ and $b = (91,2 \pm 5,0) \times 10^{-6}\ \mathrm{m^3/mol}$. These values are congruent with the literature values in [7]. The critical coefficient was calculated and we thus determined to which extent SF_6 deviates from an ideal gas. The evaluated critical coefficient read $K_\mathrm{k} = (2,79 \pm 0,65)$; hence, within the margin of error the van der Waals model is suitable for describing real sulfur hexafluoride gas.

9 Appendix

The van der Waals equation of state for real gases is shown as eq. (2). To evaluate the relation between the critical parameters and the constants a and b, one has to consider two conditions:

$$\frac{dp}{dV} = -\frac{RT_\mathrm{k}}{(V_\mathrm{k}-b)^2} + \frac{2a}{V_\mathrm{k}^3} = 0 \tag{10}$$

$$\frac{d^2p}{dV^2} = \frac{2RT_\mathrm{k}}{(V_\mathrm{k}-b)^3} - \frac{6a}{V_\mathrm{k}^4} = 0 \tag{11}$$

which leads to two separate equations:

$$RT_\mathrm{k}V_\mathrm{k}^3 = 2a(V_\mathrm{k}-b)^2 \tag{12}$$

$$2RT_\mathrm{k}V_\mathrm{k}^4 = 6a(V_\mathrm{k}-b)^3 \tag{13}$$

Dividing eq. (13) by eq. (12) gives a condition which can be solved for the critical volume V_k. The result is

$$V_\mathrm{k} = 3b. \tag{14}$$

By substituting V_k in eq. (12), the resulting critical temperature T_k follows with

$$T_\mathrm{k} = \frac{8a}{27Rb}. \tag{15}$$

Knowing V_k and T_k, substituting them in eq. (2) yields an expressions for the critical pressure p_k:

$$p_\mathrm{k} = \frac{a}{27b^2} \tag{16}$$

For the specific quantity of material $\nu = 1\,\mathrm{mol}$ we calculated the theoretically expected isotherm for an ideal gas that is described with

$$pV = \nu RT \tag{17}$$

where R is the ideal gas constant and T the temperature. We also calculated the van der Waals isotherm for a real CO_2-gas by using eq. 2 with $a = 3.658\times10^6\,\mathrm{bar\,cm^6/mol^2}$ and $b = 42,75\,\mathrm{cm^3/mol}$. Both isotherms are calculated at a temperature of 288.7 K and can be seen in fig. 8.

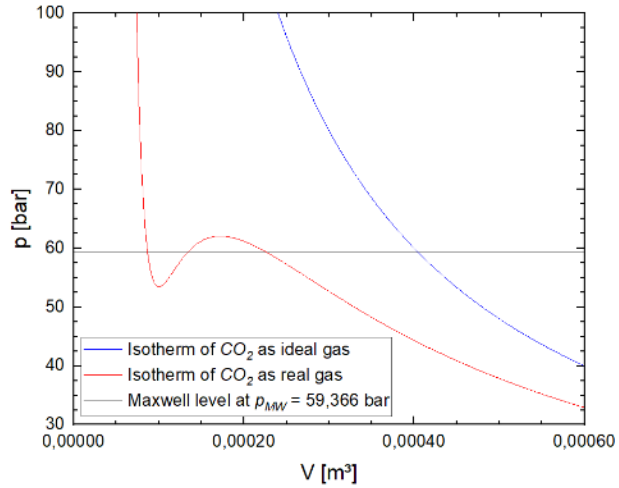

Fig. 8: Theoretical calculated isotherms for an ideal CO_2 and real CO_2.

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